

MILLINGTON-MEMPHIS, TN, USA

WMO: 723284

Lat: 35.350N

Lon: 89.867W

Elev: 320

StdP: 14.53

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93839

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.8	21.3	7.2	8.0	22.9	10.2	9.4	25.6	21.3	48.1	19.2	47.0	4.1	10	0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	99.5	80.8	97.1	79.6	93.4	77.9	83.5	96.0	81.8	93.0	80.3	91.1	6.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.7	161.3	91.3	79.0	152.5	88.8	77.2	143.1	86.7	47.4	95.4	45.7	93.0	44.0	89.9	88.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.9	15.8	13.1	10.2	104.0	2.0	3.0	8.7	106.2	7.5	108.0	6.4	109.7	4.9	111.9		
			9.1	85.3	1.6	2.4	7.9	87.0	6.9	88.3	6.0	89.7	4.8	91.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.3	40.7	43.6	55.2	63.2	70.8	79.7	81.6	80.9	73.3	62.2	52.6	42.8
	DBStd	16.70	11.09	10.39	10.14	8.36	7.32	4.72	4.57	5.50	5.74	8.27	8.77	9.23
	HDD50	977	331	231	64	7	0	0	0	0	0	6	74	264
	HDD65	3083	755	599	331	131	36	0	0	0	8	154	380	689
	CDD50	5480	43	53	227	404	646	890	981	958	700	385	153	40
	CDD65	2107	1	1	28	77	216	440	516	494	258	68	8	0
	CDH74	22610	0	6	215	721	2095	4958	5941	5605	2311	691	66	1
	CDH80	10656	0	0	28	159	773	2493	3055	2990	940	211	7	0
Wind	WSAvg	5.1	6.4	6.5	6.8	6.4	4.7	3.9	3.7	3.4	3.7	4.0	5.1	6.2
Precipitation	PrecAvg	50.8	3.7	4.2	5.1	5.7	5.2	3.8	3.6	2.8	3.7	2.8	4.8	5.4
	PrecMax	68.2	10.2	9.9	12.4	15.0	11.9	12.4	6.9	7.4	9.5	8.7	11.7	13.7
	PrecMin	33.4	0.8	1.4	0.7	1.7	0.9	0.4	1.0	0.1	0.2	0.0	1.4	0.9
	PrecStd	9.8	2.1	2.4	2.6	3.2	2.6	2.7	1.7	2.2	2.4	2.0	2.6	3.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.8	73.4	82.4	87.9	93.0	99.5	102.0	103.7	97.3	90.0	80.8	71.5
		MCWB	64.4	63.2	67.1	72.1	76.2	79.8	81.1	79.6	75.0	71.3	66.8	64.2
	2%	DB	67.6	70.2	80.6	83.9	90.4	96.9	99.2	99.9	91.4	85.4	75.1	65.9
		MCWB	61.9	61.8	65.5	68.0	74.5	78.7	82.0	80.6	74.0	69.7	63.2	61.0
	5%	DB	63.4	65.8	75.3	81.5	88.1	93.4	96.8	97.4	89.5	81.8	72.5	63.1
		MCWB	58.5	58.7	62.9	66.5	73.6	77.0	80.8	79.5	73.6	67.2	63.3	57.2
	10%	DB	59.2	62.2	72.4	79.0	84.1	91.2	92.9	93.1	85.5	77.7	69.6	57.4
		MCWB	54.5	54.4	61.0	65.7	71.6	76.5	78.9	78.7	71.6	64.9	61.1	51.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.1	66.6	69.3	73.3	79.3	82.6	85.4	85.5	79.4	76.1	71.1	65.7
		MCDB	70.1	71.7	79.5	83.8	90.3	94.9	98.1	100.3	92.5	84.5	76.4	69.9
	2%	WB	63.5	63.0	67.2	71.2	76.6	81.1	83.6	83.8	77.3	73.0	67.4	62.8
		MCDB	65.4	67.0	76.9	80.3	86.7	93.0	96.1	96.6	87.1	80.7	73.1	64.8
	5%	WB	59.4	59.6	65.4	69.3	74.9	79.6	82.1	81.8	75.7	70.7	64.6	58.3
		MCDB	62.9	65.2	73.8	77.5	84.6	91.4	93.6	92.4	84.1	77.3	70.1	62.3
	10%	WB	54.7	55.1	63.0	67.5	73.5	78.2	80.5	79.9	74.2	68.0	61.7	53.4
		MCDB	58.5	60.2	70.4	75.9	82.3	89.0	91.1	90.4	81.7	74.4	66.5	56.6
Mean Daily Temperature Range	5% DB	MDBR	18.5	19.7	21.6	22.4	21.7	21.9	20.7	22.8	22.9	23.8	21.6	19.6
		MCDBR	21.7	23.1	23.8	24.5	24.7	24.3	24.8	26.7	27.0	28.6	26.9	22.7
		MCWBR	17.8	16.4	13.3	12.3	11.3	9.6	9.8	10.4	11.9	14.0	17.2	17.6
	5% WB	MCDBR	19.7	20.3	21.3	20.3	21.2	21.9	21.9	23.2	20.5	21.8	20.2	19.7
		MCWBR	17.4	16.1	13.1	12.9	10.9	9.7	10.0	10.5	10.7	13.4	16.0	17.7
Clear-Sky Solar Irradiance	taub	0.316	0.326	0.362	0.400	0.438	0.457	0.488	0.487	0.420	0.359	0.330	0.314	
	taud	2.495	2.463	2.388	2.278	2.233	2.188	2.139	2.122	2.304	2.455	2.488	2.515	
	Ebn at Noon	282	291	288	280	270	264	255	252	266	275	273	275	
	Edh at Noon	26	30	36	42	44	47	49	49	39	30	26	24	
All-Sky Solar Radiation	RadAvg	765	930	1289	1654	1858	2068	1988	1825	1549	1206	852	654	
	RadStd	64	122	113	112	118	125	98	111	110	143	87	66	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page